Principles of Machine Learning

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Exercise Session 3: Learning theory and practical aspects in ML

3.1 The RPNI Algorithm

Let $\{aa, ab, ba\}$ be the positive examples and $\{baa\}$ the negative examples of strings over the alphabet $\Sigma = \{a, b\}$. Apply the RPNI algorithm to find a DFA that correctly labels these instances. Which regular expression does this correspond to?

Solution 3.1 First we construct the prefix tree acceptor for the positive examples. This is seen in figure 1 on the left. Double circled nodes are accepting states.

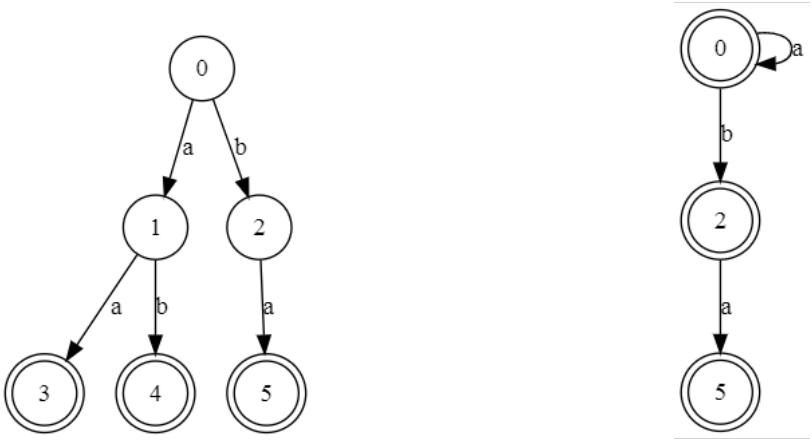


Figure 1: Prefix tree acceptor for the positive examples (left) and reduced DFA (right).

We try to merge node 1 into 0 and merge their children iteratively, yielding the graph in figure 1 on the right. Since baa is still rejected by this DFA, the merge succeeds. It is not possible to merge any further, as any further merging will accept the negative example baa. The learned regular expression is thus $a^*b?a?$ and covers the positive examples, while excluding the negative. ◇

3.2 VC-Dimension

Consider the instance space $\mathbb{R}^2$ and take as a hypothesis space the set of all hypotheses of the form "everything inside rectangle R is positive and everything outside it is negative", where R can be any axis-parallel rectangle. This means that the sides of the rectangle are parallel to the X- and Y-axis. Show that the VC-dimension of this hypothesis space is at least 3. Optionally show that the VC-dimension of this hypothesis space is at least 4.

Now consider a similar hypothesis space, where for each hypothesis that states "everything inside rectangle R is positive and everything outside it is negative" there is also a hypothesis "everything

outside R is positive and everything inside is negative". Show that the VC-dimension of this hypothesis space is at least 4.

Solution 3.2 We will show that the first hypothesis space has VC-dimension at least equal to 4. This also implies that the extended hypothesis space with "negative rectangles" also has VC-dimension larger than 4 as the VC-dimension cannot decrease by expanding the hypothesis space.

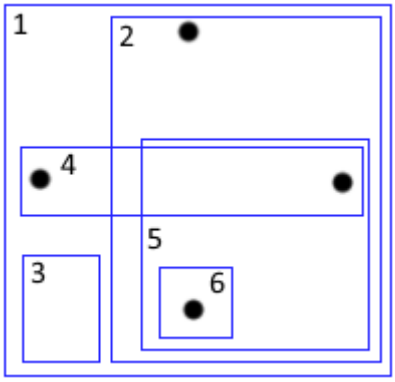


Figure 2: Illustration of how four points can be shattered by H.

To prove this lower bound, consider the 4 points $(1, 0)$, $(0, 1)$, $(-1, 0)$ and $(0, -1)$ as shown in figure 2. For any subset of these four points, it is possible to find a rectangle that covers exactly that subset. A couple of example rectangles are shown in the figure. Rectangle 1 covers all the points, rectangle 2 excludes only one point, 3 covers none, 4 and 5 cover two points, depending on whether they are adjacent or opposite each other. Lastly rectangle 6 covers only one point. Rectangles for the other possible assignments of labels can be obtained by translating or rotating one of these rectangles over a right angle. ◇

3.3 The Pizza Problem

Suppose pizzas can be composed of the following 8 ingredients: *cheese, ham, tomatoes, mushrooms, peppers, salami, olives* and *shrimp*. Further assume that the concept "pizzas that Wayne likes" can be expressed as a conjunction of "contains I" literals, with I a single ingredient. For instance, "Wayne likes all pizzas that contain peppers and salami" is a possible concept, whereas "Wayne likes all pizzas that contain peppers and no olives" is not. Show that the VC-dimension of this hypothesis space is at least 3 and at most 8.

Solution 3.3 Number the ingredients starting from 0 to 7. We represent a pizza instance by a Boolean 8-tuple, where component i represents whether ingredient i is present on the pizza, $i = 0, \ldots, 7$. Now consider the following 3 instances:

$$x = (0, 0, 0, 0, 1, 1, 1, 1),$$
$$y = (0, 0, 1, 1, 0, 0, 1, 1),$$
$$z = (0, 1, 0, 1, 0, 1, 0, 1).$$

We will show that these three instances can be shattered. Consider an arbitrary labeling $\alpha : \{x, y, z\} \to \{0, 1\}$ indicating which of the three pizzas are liked by Wayne. Then the following hypothesis exactly reproduces any possible labeling α: "the pizza contains ingredient N_α", with N_α defined as:

$$N_\alpha = 4\alpha(x) + 2\alpha(y) + \alpha(z)$$

For example, when Wayne likes pizzas x and y, but not pizza z, then the hypothesis "any pizza containing olives (ingredient $6 = 1 \cdot 4 + 1 \cdot 2 + 0 \cdot 1$)" perfectly predicts Wayne's preference. We have thus shown that $\text{VC}(H) \geq 3$.

To show that VC$(H) \leq 8$ we make use of the general bound VC$(H) \leq \log_2(|H|)$. To count the hypotheses in H we notice that each hypothesis is a disjunction of up to 8 distinct literals. Therefore there are 2^8 different hypotheses in H, because every literal can either be included or excluded from the disjunction. Filling this in in the above bound yields VC$(H) \leq \log_2(2^8) = 8$ as desired.

Alternatively: By the definition, we are looking for a set of points (i.e. pizzas x, y, z) that can be shattered for every possible labeling. In the previous exercise, you saw that not every set of points will work though: if all points are on a single line, you cannot separate them. **So we may have to choose our pizzas in a smart way.** Consider all possible labelings:

$$\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix}, \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}, \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$$

Let's say we start constructing our pizzas by being able to separate just the first three labelings, i.e. Wayne liking exactly one of the pizzas. This can be done by giving each pizza 1 ingredient:

$$x = (0, 0, 1, 0, 0, 0, 0, 0),$$
$$y = (0, 1, 0, 0, 0, 0, 0, 0),$$
$$z = (1, 0, 0, 0, 0, 0, 0, 0), .$$

Now if Wayne likes pizza x, our hypothesis becomes 'Wayne likes pizza with tomatoes' (ingredient 3), which includes x but excludes y and z. If Wayne likes exactly 2 pizzas however, these pizzas will not work anymore. We cannot use the hypothesis 'Wayne likes tomatoes or cheese', because we can only use 'and' clauses in this exercise. Luckily we still have other ingredients. By giving 2 pizzas the same ingredient (but not to the third), we can again separate them:

$$x = (0, 0, 1, 1, 1, 0, 0, 0),$$
$$y = (0, 1, 0, 1, 0, 1, 0, 0),$$
$$z = (1, 0, 0, 0, 1, 1, 0, 0), .$$

Now if Wayne likes pizza x, y (but not z) we can say that he likes pizzas with mushrooms (ingredient 4), as this topping is on pizza x and y, but not on z. These pizzas also still work for liking exactly 1 pizza, as we have not changed anything about the first three ingredients. They also work for liking 0 pizzas: if Wayne likes pizzas with olives (ingredient 7), he will not like any of these pizzas as none of them contain it. We are only missing the labeling where Wayne likes all, but this can be fixed by giving the same ingredient to all pizzas:

$$x = (0, 0, 1, 1, 1, 0, 0, 1),$$
$$y = (0, 1, 0, 1, 0, 1, 0, 1),$$
$$z = (1, 0, 0, 0, 1, 1, 0, 1), .$$

If Wayne likes pizzas with shrimp (ingredient 8), he will like all pizzas.

So now we have made 3 pizzas, and for any of Wayne's preferences (labelings) we can separate them by just using the single ingredient that matches that labeling as our hypothesis. This means that our VC dimension is at least 3. ◇

3.4 Sample Complexity

Using the concept of VC-dimension, compute a lower bound for the number of examples that may be needed to train a 2-input perceptron such that with 90% certainty it learns a hypothesis with true error smaller than 5%.

Hint: The first step is to find a lower bound for the VC-dimension of a perceptron. Then, look in the slides for a formula linking VC-dimension to number of required samples.
Hint 2: A 2-input perceptron is equivalent to a 2D linear separator.

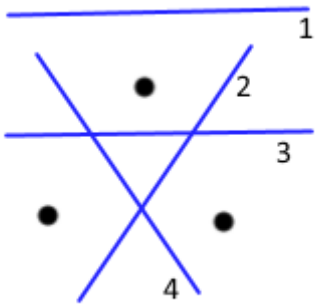


Figure 3: Three points being shattered by a linear separator.

Solution 3.4 The VC-dimension of a perceptron is at least 3, as it is essentially a linear separator. Figure 3 illustrates this. Notice that for every labeling of the points, one can find a line such that points on opposite sides of the line have opposite labels.
The next step is to argue that the upper bound of this VC-dimension is also 3. For this we refer to slide 56: "The VC-dimension of linear separators in 2D is 3: there does not exist a set of 4 points that is shattered by this hypothesis space". Which means that VC$(H) = 3$.

Using inequalities from the slides yields

$$|T| \geq \frac{1}{\epsilon}\left[4\log_2\left(\frac{2}{\delta}\right) + 8\text{VC}(H)\log_2\left(\frac{13}{\epsilon}\right)\right] = 20(4\log_2(20) + 8 \cdot 3 \cdot \log_2(260)) \approx 4196.491.$$

Because the first $\geq$ still holds, this number represents a lower bound and the number of samples must be at least 4197. ◇

3.5 Feature Selection

Assume we want to classify instances into three classes c_1, c_2 and c_3. Our original input has three features a_1, a_2 and a_3 of which we want to keep only two. Feature a_1 is similar for instances of classes c_1 and c_2, but different for c_3 instances. Feature a_2 is similar for instances of classes c_2 and c_3, but different for c_1 instances. Instances of all three classes have similar a_3. Let f_i be the fraction of instances of class c_i in the training dataset. We know that $f_1 > f_2 > f_3$.

When performing greedy forward feature selection, which feature will get selected first? Which second? Why?

Solution 3.5 If we were to select a_1 first, we can distinguish c_3 instances and will correctly classify them, assuming we have a capable learning algorithm. If the prediction instance turns out to be of class c_1 or c_2 however, we cannot distinguish them, so the best we can do is predict the majority class c_1. Therefore this feature will give an accuracy of $f_1 + f_3$ on the training set. Using a similar reasoning, with the classes swapped, we see that a_2 will have an accuracy of $f_1 + f_2$ and a_3 of f_1. Since $f_2 > f_3$, it is best to choose a_2 first.

For the second feature, we still need to be able to distinguish c_2 and c_3 instances, which is done by a_1, but not a_3. Therefore we will choose a_1 as a second feature.

It is easy to see why a_3 is not selected; as it is similar for all instances, it contains no new information and is not useful for the classification task at hand. ◇

3.6 Fatigue Prediction

Take a look at this paper[1] which can be found at `https://dl.acm.org/citation.cfm?id=3219864`. The aim of this research is to predict using machine learning techniques how fatigued runners are after performing with the aim of preventing injuries.

For this purpose, five accelerometers were attached to the runners, one at the back, one at each wrist and one at each leg. The data of all these sensors can be combined to predict the runner's fatigue. It is also possible to fuse this data together before using it to predict. Furthermore some general statistical features and some more advanced sport science features are calculated on these temporal data.

Read the paper and try to think about the following questions:

1. Which sensor locations are most useful?
2. Does fusing data affect prediction and run time performance?
3. Are the advanced sport science features useful?

Solution 3.6 The answers to the questions are respectively:

1. The arm sensor is comparable or sometimes even better than the leg sensor, but is more practical and thus the preferred sensor.
2. Since the acceleration of the different body parts are strongly correlated when running, it suffices to keep track of only one combined acceleration. This can slightly decrease performance, as extra information is discarded, but can increase run time performance as only one signal has to be processed.
3. The advanced features do slightly improve the predictive performance, but are also computationally expensive to calculate and might not be worth the extra computations compared to the simpler statistical features.

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3.7 Feature Selection

During the lectures, we have discussed multiple strategies for doing feature selection. In this practical exercise, you'll explore the difference between three strategies: Principal Component Analysis, Single Variable Classifier-based Feature Selection and Backward Feature Selection.

Download the 'feature_selection_assignment' notebook and complete the gaps. Why do you think one feature selection method performed better than the others? How does the nature of the dataset influence the effectiveness of each method?

Solution 3.7

- PCA reduces dimensionality by preserving variance but loses the original features. Hence, features can no longer be interpreted. It's an unsupervised approach and does not take class labels into account. Therefore, it does not necessarily lead to improvement of a classifier's predictive performance.
- Single Variable Classifier-based selection highlights important features based on how much they reduce classification error in a simple model, and is interpretable. It is fast and one can select features according to the metric that they're trying to optimize for (i.e., accuracy in this exercise). However, useful combinations of features are not being looked at.

[1] Fatigue Prediction in Outdoor Runners Via Machine Learning and Sensor Fusion; Tim Op De Beéck, Wannes Meert, Kurt Schütte, Benedicte Vanwanseele, Jesse Davis; 2018

- Backward selection removes one feature at a time and retrains the model. It's more computationally expensive but can provide an optimal feature set, balancing performance and interpretability.

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3.8 Comparing Models

Every ML model has its own strengths and weaknesses. Often, it is not obvious which model would work best on a certain dataset. Hence, it is important to be able to compare models. In this exercises, you'll perform a paired t-test to compare two machine learning models using 10-fold cross validation. Download the 't_test_assignment' notebook and fill in the gaps.

Answer the following questions:

- Based on the p-value, can we reject the null hypothesis that both models have the same performance? Discuss what a p-value lower than 0.05 would mean in this context.
- Why is a paired t-test used in this case, rather than an independent t-test?
- How does cross-validation help ensure the reliability of the comparison between the two models?

Solution 3.8

- The null-hypothesis can be rejected if the p-value is smaller than the level of significance. Often, this value is set to 0.05. In that case, we can reject the null-hypothesis if the p-value < 0.05. However, it is prudent to set the significance level to values even smaller. The meaning of the p-value can be explained as the probability of finding results that are as extreme as the actually observed results, assuming that the null-hypothesis is true.
- A paired t-test is used because the same dataset is used to evaluate both models. In a paired t-test, we account for the fact that the performance scores for both models come from the same data splits (in this case, the same 10-fold cross-validation splits). This creates a dependency between the scores, which is why we "pair" them.

  If we used an independent t-test, it would assume the data points (the cross-validation scores) from both models are independent, which is not the case here. A paired t-test is more appropriate because it directly compares the difference in performance within the same conditions (same data split), reducing variability and improving the sensitivity of the test.
- Cross-validation improves the reliability of the comparison by reducing bias and variance in the model evaluation process. By using 10-fold cross-validation each model is trained multiple times on different data splits. The multiple scores for each model allow the use of statistical tests, improving the reliability of the comparison.

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